

Recurrence Relation.

It is an eqn that recursively defines a sequence where the next term is a fn of the previous term.

eg: i) $2, 2^2, 2^3, \dots, 2^n, \dots$

$$\{s_n\} = \{2^n\}$$

$$S_n = 2S_{n-1}, \quad n \geq 2, \quad S_1 = 2 \rightarrow \text{I.C.}$$

ii) $3, 7, 11, \dots$

$$S_n = S_{n-1} + 4, \quad n \geq 2, \quad S_1 = 3$$

iii) $1, 1, 2, 3, 5, \dots$

$$S_n = S_{n-1} + S_{n-2}, \quad n \geq 3, \quad S_1 = S_2 = 1$$

Linear RR with const. co-eff.

$$c_0 a_n + c_1 a_{n-1} + \dots + c_k a_{n-k} = f(n),$$

Where c_i - are constants.

order: diff b/t highest and lowest subscript of a .

Homogeneous if $f(n) = 0$

Non Homogeneous if $f(n) \neq 0$.

Generating Functions.

$$a_0, a_1, a_2, \dots, a_k, \dots$$

$$G(x) = a_0x^0 + a_1x^1 + a_2x^2 + \dots + a_kx^k + \dots$$
$$= \sum_{k=0}^{\infty} a_k x^k$$

eg. 1. $1, 1, 1, \dots$

$$G(x) = 1 + x + x^2 + \dots$$

closed form

$$G(x) = \frac{1}{1-x} \quad |x| < 1$$

2. $2, 2, 2, \dots$

$$G(x) = 2 + 2x + 2x^2 + \dots$$

$$G(x) = \frac{2}{1-x}$$

3. $1, -1, 1, -1, \dots$

$$G(x) = \frac{1}{1+x}$$

4. $1, 2, 3, 4, 5, \dots$

$$G(x) - xG(x) = \frac{1}{1-x}$$

5. $0, 1, 2, 3, \dots$

$$G(x) = \frac{1}{(1-x)^2}$$

6. $1, a, a^2, a^3, \dots$

$$G(x) = \frac{1}{1-ax}$$

7. $1, -a, a^2, -a^3, \dots$

$$G(x) = \frac{1}{1+ax}$$

8. $0, 0, 1, 1, 1, \dots$

$$G(x) = 0 + 0x + x^2 + x^3 + \dots$$

$$= x^2(1 + x + x^2 + \dots) = \frac{x^2}{1-x}$$

9. $1, 1, 0, 1, 1, \dots$

$$G(x) = 1 + x + 0 \cdot x^2 + x^3 + x^4 + \dots$$

$$= \frac{1}{1-x} - x^2$$

10. $1, 0, -1, 0, 1, 0, -1, \dots$

$$G(x) = 1 - x^2 + x^4 - x^6, \dots = \frac{1}{1+x^2}$$

11. $a_k = 2 + 3k$

$$G(x) = \frac{2}{1-x} + \frac{3x}{(1-x)^2}$$

12. $a_k = 4^k$ $1 + 4x + 4^2x^2 + 4^3x^3 + \dots$

$$G(x) = \frac{1}{1-4x}$$

13. $a_k = (-2)^k$ $1 - 2x + (-2)^2x^2 + (-2)^3x^3 + \dots$

$$G(x) = \frac{1}{1+2x}$$

	a_k	$G(x)$		a_k	$G(x)$
1	1	$\frac{1}{1-x}$		a^k	$\frac{1}{1-ax}$
	$(-1)^k$	$\frac{1}{1+x}$		$(-a)^k$	$\frac{1}{1+ax}$
	c	$\frac{c}{1-x}$		k^2	$\frac{x(1+x)}{(1-x)^3}$
	$k+1$	$\frac{1}{(1-x)^2}$		$k a^k$	$\frac{ax}{(1-ax)^2}$
	k	$\frac{x}{(1-x)^2}$			

Solve:

$$a_r - 2a_{r-1} - 3a_{r-2} = 0, \quad a_0 = 3, \quad a_1 = 1$$

Let $G(x) = \sum_{r=0}^{\infty} a_r x^r$

Solu $\sum_{r=2}^{\infty} a_r x^r - 2 \sum_{r=2}^{\infty} a_{r-1} x^r - 3 \sum_{r=2}^{\infty} a_{r-2} x^r = 0$

$$G(x) - a_0 - a_1 x - 2x(G(x) - a_0) - 3x^2 G(x) = 0$$

$$G(x)(1 - 2x - 3x^2) = a_0 + a_1 x - 2x a_0$$

$$= 3 + x - 6x = 3 - 5x$$

$$G(x) = \frac{3 - 5x}{(1 - 2x - 3x^2)} = \frac{5x + 3}{3x^2 + 2x - 1}$$

$$= \frac{5x + 3}{(1 + 3x)(1 - 2x)}$$

$$3x(1+x) - (x+1) = 3x^2 + 3x - x - 1$$

$$= \frac{5x+3}{(3x-1)(1+x)} = \frac{A}{3x-1} + \frac{B}{1+x}$$

$$5x+3 = A(1+x) + B(3x-1)$$

$$A+B=3 \quad \text{const}$$

$$A+3B=5 \quad \text{coeff } x$$

$$-4B = -8 \quad B = 2 \Rightarrow A = 1$$

$$g(x) = \frac{1}{-1-3x} + \frac{2}{1+x} = \frac{1}{1-3x} + \frac{2}{1+x}$$

$$a_r = 2(-1)^r + 3^r$$

Solve

$$a_r - a_{r-1} - 2a_{r-2} = 0 \quad a_0 = 0 \quad a_1 = 1$$

Soln

$$g(x) = \frac{x}{1-x-2x^2}$$

$$= \frac{x}{(1-2x)(1+x)}$$

$$= \frac{1/3}{1-2x} + \frac{-1/3}{1+x}$$

$$\therefore a_r = \frac{1}{3}(2)^r - \frac{1}{3}(-1)^r$$

$$-(2x^2+x-1)$$

$$-(2x^2+2x-x-1)$$

$$-(2x(x+1)-(x+1))$$

$$-(2x-1)(x+1)$$

Solve $a_n = 3a_{n-1} + 2$, $a_0 = 1$

Soln $\sum_{n=1}^{\infty} a_n x^n = 3 \sum_{n=1}^{\infty} a_{n-1} x^n + 2 \sum_{n=1}^{\infty} x^n$

$$G(x) - a_0 = 3x G(x) + 2 \left(\frac{1}{1-x} - 1 \right)$$

$$G(x) - 1 = 3x G(x) + 2 \frac{x}{1-x}$$

$$G(x)(1-3x) = \frac{2x}{1-x} + 1 = \frac{1+x}{1-x}$$

$$G(x) = \frac{1+x}{(1-x)(1-3x)} = \frac{-1}{1-x} + \frac{2}{1-3x}$$

$$a_n = -1(1) + 2 \cdot 3^n$$

Solve $a_{n+2} - 2a_{n+1} + a_n = 2^n$, $a_0 = 2$, $a_1 = 1$

Soln $\sum_{n=0}^{\infty} a_n x^n$

Let $G(x) = \sum_{n=0}^{\infty} a_n x^n$

$$\sum_{n=0}^{\infty} a_{n+2} x^n - 2 \sum_{n=0}^{\infty} a_{n+1} x^n + \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} 2^n x^n$$

$$\frac{1}{x^2} (G(x) - a_0 - a_1 x) - 2 \frac{1}{x} (G(x) - a_0) + G(x) = \frac{1}{1-2x}$$

$$G(x) - 2 - x - 2x(G(x) - 2) + x^2 G(x) = \frac{x^2}{1-2x}$$

$$G(x)(x^2 - 2x + 1) - 2 - x + 4x = \frac{x^2}{1-2x}$$

$$G(x) (1-x)^2 = 2 - 3x + \frac{x^2}{1-2x}$$

$$G(x) = \frac{2}{(1-x)^2} - \frac{3x}{(1-x)^2} + \frac{x^2}{(1-2x)(1-x)^2}$$

$$\frac{x^2}{(1-2x)(1-x)^2} = \frac{A}{1-2x} + \frac{B}{1-x} + \frac{C}{(1-x)^2}$$

$$x^2 = A(1-x)^2 + B(1-2x)(1-x) + C(1-2x)$$

$$\text{Let } x=1 \quad 1 = -C \Rightarrow C = -1$$

$$x=0 \quad 0 = A+B+C \Rightarrow B=0$$

$$x=1/2 \quad \frac{1}{4} = \frac{A}{4} \Rightarrow A=1$$

$$\therefore G(x) = \frac{2}{(1-x)^2} - \frac{3x}{(1-x)^2} + \frac{1}{1-2x} + \frac{0}{1-x} + \frac{-1}{(1-x)^2}$$

$$= \frac{1}{(1-x)^2} - \frac{3x}{(1-x)^2} + \frac{1}{1-2x}$$

$$a_n = n+1 - 3n + 2^n$$

$$= 1 - 2n + 2^n$$